Scutoid Tissues in Four Dimensions

the boundary of every simple 4-polytope is a closed scutoid tissue whose moves are counted by its h-vector; the Swap Rule of close packing below s = 1/4, whose walls are rosette exchanges

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DICTO: ideas, names, constructions and direction.
Computation, verification and drafting with Claude (Anthropic).

Version 3 adds references: the 600-cell family (Coxeter; Conway and Guy; Conway, Burgiel and Goodman-Strauss; DICTO #63, #65, #76) and the scutoid follow-ups that count exchanges (Gómez-Gálvez et al. 2021, 2022), each checked against its publisher's record, with a sentence placing Theorem 1 beside the biological count; and a gallery search for the figures, credited in Figure 3 (views centred on a truncated icosahedron are known; we found no view in adjacency shells, cut open or pulled apart). No result changes.

Version 2 adds, after an external review: the terminology (scutoid tissue, 4D scutoid); Theorem 1′ for d ≥ 4 with the table of indices; the cells present at the start and the end; what is classical and what is new; the g-theorem in tissue terms; the four-layer condition defined; Theorem 3 as a computer-assisted proof with a Verification section, re-run from a clean checkout of the repository with identical results; interpretations labelled; a reference list; three figures from the app's Bitruncated 600-cell world. No result changes.

Version 1. Theorems 1, 1′, 2 and 4 and Lemma 5 are proved here; their counts are also confirmed by computer sweeps (floating point), each predicted before counting. Theorem 3 is an exact computer-assisted proof for every layer spacing in [1/8, 1/4] in the six stackings that contain all sixteen four-layer contexts; for other stackings it assumes that a piece is fixed by the four layers round its gap; the region below $\lambda = -9s/8$ and $s < 1/8$ are open. Checked before release by four automated referee passes and one independent attack (reports and judgements kept with the study data).

A scutoid is a cell between two surfaces whose outline changes on the way through: two cells that touched below stop touching, and two others start. Gómez-Gálvez and co-workers found the shape in curved living tissue in 2018; they called the change an apico-basal transition and compared it to the T1 transition of foams and cell sheets (in a 2D cell pattern a T1 is the classical 2-2 flip of the dual triangulation: Weaire and Rivier; Weaire and Hutzler). Later work by the same group counts these exchanges: in epithelial tubes the cells' 3D connectivity, and with it the number of apico-basal exchanges, grows with the tissue's thickness and curvature by a logistic law (Gómez-Gálvez et al. 2022; review: Gómez-Gálvez et al. 2021). Theorem 1 below is the polytope counterpart: there the number of exchanges is fixed by the h-vector. We find such tissues where one would not look: on the boundary of a four-dimensional polytope.

Slice a 4-polytope with a moving hyperplane; its 3-dimensional cells cross the slice as polygons tiling the slice's surface, and as the hyperplane moves the polygons trade neighbours. Each cell, followed through the sweep, is a flat-faced scutoid in the 2018 sense, and the boundary is a closed scutoid tissue. For simple 4-polytopes the number of neighbour exchanges is fixed by the h-vector, in every direction; for the polytope halfway between the 600-cell and the 120-cell they balance exactly between its two kinds of cell. Then one dimension up (Section 3): the holes of close packing, their power cells stacked over a weight λ, are 4-dimensional cells whose 3D slices exchange neighbours at the walls of the Swap Rule (#61), there in symmetric clumps of the kind biologists call rosettes.

Terminology. Two constructions carry the word here, and they are kept apart. A *scutoid tissue* (Section 1) lives on the boundary of a simple 4-polytope: its cells are the polytope's 3-dimensional cells, and their sections by a moving 3-space are polygons that exchange neighbours; each cell followed through the sweep is a scutoid in the 2018 sense, and a *scutoid swap* is one such exchange. A *4D scutoid* (Section 3) is a single 4-dimensional cell, a hole's power cell stacked over the weight λ, whose 3D sections exchange neighbours. Both are convex with flat faces, unlike the biological scutoid, whose faces are curved and which need not be convex.

1. The boundary of a simple 4-polytope is a scutoid tissue

A 4-polytope is *simple* if each corner lies in exactly four cells (equivalently, four edges). Take a generic height, a linear function with different values at all corners. Slicing at height t gives a 3-polytope whose faces are the cross-sections of the cells that cross the hyperplane; two cells are neighbours in the slice when their shared 2-face crosses it. At a simple corner the faces through it correspond to the subsets of its four edges, and a face has the corner as its top iff all its edges there go down.

Theorem 1. In a generic sweep of a simple 4-polytope the tiling of the slice by the cells changes only when the hyperplane passes a corner, and then in exactly one of three ways, according to the number i of the corner's edges that go down: i = 1, one cell is born (a small triangle appears); i = 2, a **scutoid swap**: one pair of cells stops touching and one pair starts, nothing else; i = 3, one cell dies. The numbers of births, swaps and deaths are h_1, h_2 and h_3, the middle entries of the polytope's h-vector, the same for every generic direction.

Proof. Between corner heights the slice keeps its combinatorics. At a corner v let $e_1, \dots, e_4$ be its edges and F_j the cell not containing e_j. Just below v the slice cuts the downward edges, just above the upward ones. i = 1 (e_1 down): above v the slice cuts e_2, e_3, e_4, and the cell containing all three, F_1, appears as a small triangle (v is its lowest corner). i = 2 (e_1, e_2 down): below v the slice's local edge joins the crossings of e_1 and e_2 and lies in $F_3 \cap F_4$, the 2-face spanned by e_1 and e_2; v is its top, so above v the two cells no longer share a face crossing the hyperplane (two cells of a polytope share at most one 2-face); symmetrically $F_1 \cap F_2$, spanned by e_3, e_4, has v as its bottom and F_1, F_2 begin to touch. No other face through v has v as top or bottom, so nothing else changes. i = 3 is i = 1 upside down. The number of corners with i downward edges is h_i, independent of the generic height (the Morse-type count of the h-vector, with $h_i = h_{4-i}$; McMullen; Ziegler, Lectures on Polytopes, Lecture 8; Buchstaber–Panov, Toric Topology, ch. 1). ∎ In the dual triangulation of the slice's boundary (one vertex per cell) the three events are the bistellar flips 1-3, 2-2 and 3-1 (Pachner; Lickorish); the theorem is classical in that language.

Classical and new. Classical: that the number of corners of index k is h_k for every generic linear function (McMullen 1970; the line shellings of Bruggesser and Mani 1971; Ziegler, Lecture 8), the Dehn–Sommerville relations, and the bistellar moves. New here: the reading of the three indices as the birth, scutoid swap and death of the cells of a tissue, with h_2 counting the swaps; the counts of Theorem 2; the balance law, Theorem 4; and Sections 2 and 3.

Remark (the g-theorem in tissue terms). For a simple 4-polytope the g-theorem (McMullen's conditions, proved by Stanley 1980 and Billera and Lee 1981) gives $g_2 = h_2 - h_1 \geq 0$ and $g_2 \leq C(h_1, 2)$. So every generic sweep of a simple 4-polytope has at least as many swaps as births, and at most births + C(births, 2): for the three polytopes of Theorem 2, 366 ≥ 116, 966 ≥ 716 and 2166 ≥ 716.

Theorem 1′ (every dimension d ≥ 4). In a generic sweep of a simple d-polytope, d ≥ 4, two facets touch in the slice exactly when their common ridge crosses the hyperplane. Passing a corner with i downward edges is the bistellar $i \to (d - i)$ move on the dual triangulation of the slice's boundary, and it changes the facets' contacts only for i = 1 (a facet is born), i = 2 (one contact is gained), i = d − 2 (one contact is lost) and i = d − 1 (a facet dies). For d = 4, i = 2 = d − 2 and the gain and the loss happen together (the swap of Theorem 1); for d = 5 they are separate moves at i = 2 and i = 3; for d ≥ 6 the corners with $3 \leq i \leq d - 3$ change no facet contact (they move lower faces only). The number of corners of index i is h_i.

Proof. A ridge $F_a \cap F_b$ through v is spanned by the d − 2 edges other than e_a, e_b; it has v as its top iff they all go down (so i = d − 2 and {a, b} are the up edges), as its bottom iff they all go up (i = 2); otherwise it crosses on both sides. ∎ *Confirmation* (floating point, predictions first, three random directions each): the 5-cube, h = (1, 5, 10, 10, 5, 1): 5 births, 10 contacts gained, 10 lost, 5 deaths; the 5D permutohedron (the 720 orderings of 1, …, 6), h = the Eulerian numbers (1, 57, 302, 302, 57, 1): 57, 302, 302, 57; the 4D permutohedron, (1, 26, 66, 26, 1): 26 births, 66 swaps, 26 deaths. **No regularity is needed:** three random simple 4-polytopes (polars of hulls of 24, 36, 54 random points near the 3-sphere, no symmetry) gave exactly the predicted (20, 57, 20), (32, 103, 32), (50, 142, 50). *Remark.* At a corner of a non-simple polytope the moves clump; a generic small shift of the facet hyperplanes makes any polytope simple and splits each clump into births, swaps and deaths.

Remark (the indices). In a simple d-polytope: index 1, a facet is born; index 2, a contact is gained; index d − 2, a contact is lost; index d − 1, a facet dies. Only for d = 4 do the gain and the loss share an index, and make the swap; for d = 5 they separate (2 and 3); for d ≥ 6 the indices 3, …, d − 3 touch only lower faces. For d = 3 two of them coincide the other way: index 1 is both a birth and a contact lost, and index 2 both a contact gained and a death (the slice polygon gains a side where two sides stopped meeting, and loses one where two begin to meet).

Theorem 2 (the polytopes between the Queens). On the path of uniform truncations from the 600-cell (the White Queen, #63) to the 120-cell (the Red Queen, #65) (the stages between: #76; the uniform 4-polytopes: Conway and Guy 1965; Coxeter, Regular Polytopes), the 120-cell, the truncated 120-cell and the bitruncated 600-cell are simple, with (births, swaps, deaths) = (116, 366, 116), (716, 966, 716) and (716, 2166, 716) in every generic sweep.

Proof. Their vertex figures are tetrahedra; $h_1 = f_3 - 4$ and $h_2 = f_0 - 2 - 2h_1$ follow from the Dehn–Sommerville relations; then Theorem 1. *Confirmation:* computer sweeps along generic directions give the predicted counts (the bitruncated 600-cell along nine directions). The four cells at the lowest corner are in the slice from the start and four remain at the top corner, so births + 4 = deaths + 4 = cells (120 = 116 + 4; 720 = 716 + 4). ∎

Lemma 5 (the truncated 600-cell has no pure swap). At every corner of the truncated 600-cell, in every direction, at least three contacts change at once.

Proof. Six cells meet at each corner (one icosahedron and five truncated tetrahedra); the vertex figure is a pentagonal pyramid, whose graph is the wheel W_5 with 10 edges. A pure swap needs the down/up split of the vertex figure's vertices to leave exactly one edge inside each side; but the largest cut of W_5 has 7 edges (the hub with two non-adjacent rim vertices), so at least three edges lie inside the two sides, each a contact that is lost or gained. ∎ In five computer sweeps the moves were $1 \leftrightarrow 3$, $2 \leftrightarrow 0$ and $1 \leftrightarrow 0$ clumps.

Interpretation. Biologists know the same contrast in living tissue: a T1 transition involves four cells at a point, while a *rosette*, five or more cells meeting at one point and resolving together, reshuffles several neighbours at once (Blankenship, Backovic, Sanny, Weitz and Zallen 2006). In that language the truncated 600-cell, with its groups of five truncated tetrahedra round each icosahedron's corner, is a rosette tissue, and the bitruncated 600-cell, one stage on, a T1 tissue.

polytope	corners	cells	cells per corner	births	scutoid swaps	deaths
120-cell (Red Queen)	600	120 dodecahedra	4	116	366	116
truncated 120-cell	2400	120 truncated dodecahedra + 600 tetrahedra	4	716	966	716
bitruncated 600-cell (the midpoint)	3600	120 truncated icosahedra + 600 truncated tetrahedra	4	716	2166	716
truncated 600-cell	1440	120 icosahedra + 600 truncated tetrahedra	6	rosette moves only: ≥ 3 contacts change at every corner (Lemma 5)		

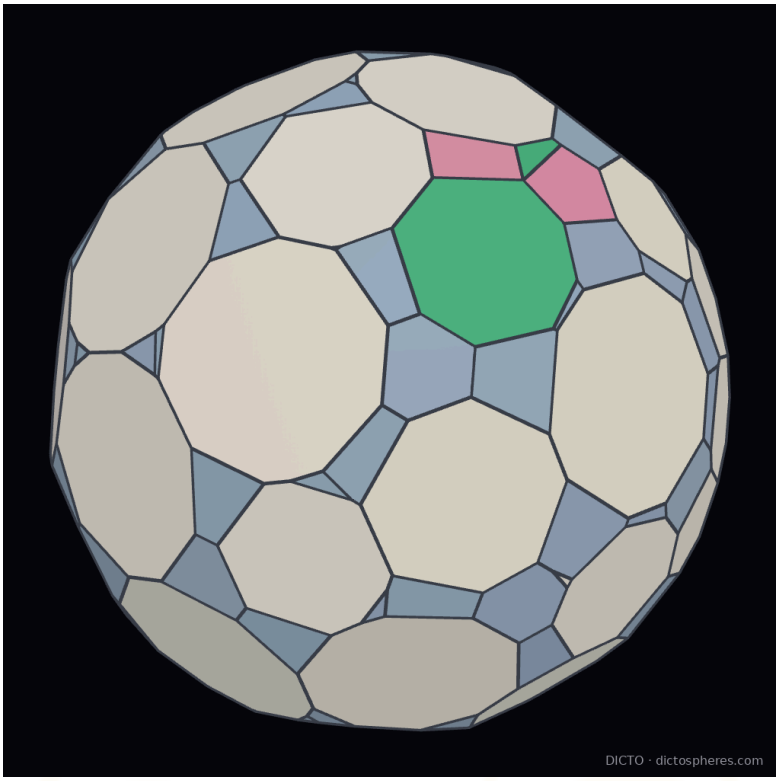


Figure 1. A scutoid tissue at work: the bitruncated 600-cell sliced by a generic 3-space at its 1083rd swap of 2166. The slice's surface is tiled by the sections of 50 truncated icosahedra (ivory) and 107 truncated tetrahedra (blue); the two cells in rose let go of each other and the two in green begin to touch (the app's Bitruncated 600-cell world, Sweep).

In symmetric directions many corners share a height and the events come in batches; along cell-first the bitruncated 600-cell's pairs that let go and pairs that meet are equal in total (1782 each) and at every height where no cell appears or disappears.

Theorem 4 (the balance law). Let a simple 4-polytope have its cells in two kinds, W and R, with 2 W + 2 R cells at every corner. Then in every generic sweep the swaps that exchange a W–W contact for an R–R contact and those that exchange R–R for W–W are equally many, (R–R faces) − (R cells) + 1, and every other swap exchanges one W–R contact for another. For the bitruncated 600-cell (W: its 600 truncated tetrahedra, the 600-cell's cells cut back; R: its 120 truncated icosahedra, in the places of the 120-cell's cells): 601 each way and 964 mixed.

Proof. With 2 W and 2 R at a corner, the pair that lets go and the pair that meets are complementary: W–W ↔ R–R or W–R ↔ W–R. Every R–R contact appears exactly once in a sweep: at the first corner (one), at the birth of an R cell after the start (one each, R cells − 2 of them) or by a W–W → R–R swap; so these swaps number (R–R faces) − (R cells − 2) − 1; by deaths the same count goes the other way. Bitruncated 600-cell: 720 − 118 − 1 = 601; the triangles check it, 1200 = 601 + 598 + 1, and 2166 − 1202 = 964. Confirmed in nine directions. ∎

Interpretation. The Queens are the two ends of a channel and this layer, where their cells meet, is where they communicate, equally in both directions.

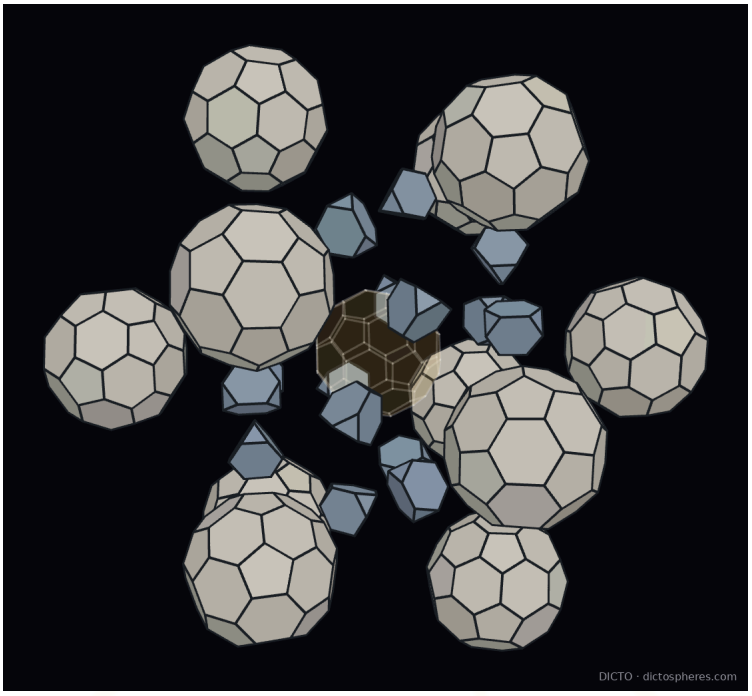


Figure 2. The bitruncated 600-cell in shells: round the gold core, its first shell of 32 cells (12 truncated icosahedra, 20 truncated tetrahedra), pulled apart so each cell floats free (the app, Shells with Pull apart).

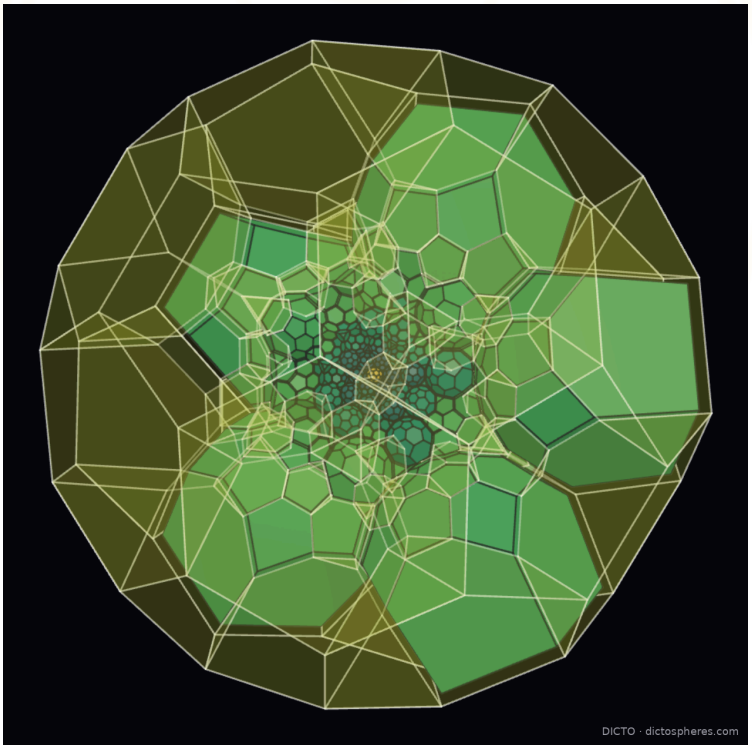


Figure 3. All 720 cells in their six shells round the core (1, 32, 142, 242, 232, 71 cells), cut open through the centre like an onion (the app, Shells, Cut open). Views centred on one truncated icosahedron are known (a Schlegel diagram by T. Ruen with R. Webb's Stella4D, 2010; H. S. Teoh's stage-by-stage build by 4D height); we found no published view of its cells in adjacency shells, cut open or pulled apart.

2. The Swap Rule below s = 1/4

Setup (#61, The Swap Rule). Layers of balls (spacing 1 in a layer) are stacked in a word in A, B, C with layer distance d, $d^2 = (2/3)s$ (s = 1: close packing). Between two layers lie octahedral and tetrahedral holes; space is divided among the holes by power distance $|x - p|^2 - w$, w = 0 for an octahedral hole and w = λ for a tetrahedral one, with λ in #61's range $-1/4 - s/8 \le \lambda \le 3s/8$ (for $s \le 1/4$). A layer is c if its two neighbours differ, h if they are equal. A hole's piece in the gap between layers k and k + 1 is keyed by its role (octahedral, or one of the two tetrahedral kinds) and the types of the four layers k − 1, k, k + 1, k + 2; the types of the two inner layers are (t | t′), the two outer ones the context. V(t | t′) is the piece's volume with the context held fixed. #61 proved V(c|h) + V(h|c) = V(c|c) + V(h|h) when $s \ge 1/4$.

The four-layer condition. A piece's *context* is its role (octahedral, or one of the two tetrahedral kinds) together with the types, c or h, of the four layers k − 1, k, k + 1, k + 2 round its gap. A stacking satisfies the *four-layer condition* at (s, λ) when each of its pieces is determined, up to a motion, by its context. The six stackings below contain all sixteen contexts, and their pieces are computed directly; for any other stacking Theorem 3 assumes the condition. Below $\lambda = -9s/8$ it fails: a piece reaches two gaps.

Theorem 3 (computer-assisted proof). For every s in [1/8, 1/4], in the six stackings ABC, AC, ACBC, ACABC, ACBCAC, ACBABC (which contain all sixteen four-layer contexts), and in every stacking that satisfies the four-layer condition, the Swap defect D = V(c|h) + V(h|c) − V(c|c) − V(h|h) of each role is the same for all four contexts and equals: for $-9s/8 \le \lambda \le -(4 - s)/24$ (when s > 1/7), $\delta_{tet} = -(1 - 4s)^3/(54s)$, $\delta_{oct} = (1 - 4s)^2(72\lambda + 8 + 13s)/(108s)$; for $\max(-9s/8, -(4 - s)/24) \le \lambda \le -5s/8$, $\delta_{oct} = (8\lambda + 5s)^3/(64s)$, $\delta_{tet} = -(8\lambda + 5s)^2(8\lambda + 2 - 3s)/(64s)$; and D = 0, **the Swap Rule holds**, from $\lambda = -5s/8$ to 3s/8 (a further wall at −s/8 leaves D = 0). Below −9s/8 a piece reaches two gaps and the four layers do not fix it (open).

Proof (exact, computer-assisted). s enters only the vertical metric, so every power-diagram plane has a normal linear in s and an offset linear in s and λ; for fixed s only the right-hand sides carry λ, so corners and slacks are affine in λ, and a slack is non-negative on a cell of the map iff it is on the cell's two bounding walls: one-variable rational functions of s. On each cell of the map, built at one sample point for the six words, every corner was re-solved symbolically over Q(s) and every plane's slack proved non-negative on the whole cell (Bernstein sign tests over the s-interval, exact root counting otherwise; 35,561,184 exact sign checks); the volumes and D(s, λ) agree with the closed forms as rational identities. *Neighbourhood.* The cells were built from all sites within squared distance 24 (every plane's slack proved non-negative on the whole cell). No farther site can cut a cell: for any point x of a hole's cell, the octahedral holes of the nearest gap form a triangular lattice of spacing 1 at vertical offset ≤ d/2, so the nearest of them, p_k, has $|x - p_k|^2 \le 1/3 + 1/24 = 3/8$, and hence $|x - p_i|^2 \le 3/8 + |\lambda| < 0.67$: every cell lies within $\rho < 0.82$ of its site, and $(\sqrt{24} - 0.82)^2 - 0.82^2 > 15.9$ exceeds every weight difference $|\lambda| \le 0.29$. The walls were first found exactly along λ (the Hopscotch walk: each wall is a root where a corner meets a plane) at six spacings. ∎

Verification of Theorem 3

Domain. s in [1/8, 1/4], λ from −9s/8 to 3s/8 (from −(4 − s)/24 when s ≤ 1/7), the six stackings ABC, AC, ACBC, ACABC, ACBCAC, ACBABC, every site within squared distance 24; open: λ below −9s/8 and s below 1/8. *Exact and floating-point steps.* Floating point is used only to propose each cell's structure at one sample point; every corner is then re-solved over Q(s), every slack is a rational

function, and every sign is decided exactly (Bernstein bounds over the s-interval, exact root counting otherwise). No floating-point step decides a sign; the closed forms are checked as rational identities. *Environment.* Python 3.14.4 with SymPy 1.14.0 (DomainMatrix over Q(s)), Ubuntu 26.04 Linux. *Certificates.* The five cell logs data/cells2-D24/{b, c1, c2, d1, d2}.txt in studies/2026-11-09-b1-walls (MD5 f08927c4..., f3bbe3cf..., be695d8f..., 2b7beddb..., 9b1832d0...), each ending "FINAL ok (cell X proved for all s in ..., neighbourhood $D^2 = 24$)". *Re-run.* From a clean checkout: `cd studies/2026-11-09-b1-walls/scripts && D2=24 python -u cells61.py`; about 45 minutes with the five cells in parallel; it ends "FINAL ok". This was done on 11 November 2026 from a clean copy of the repository, and every cell log came out identical to the original apart from timings.

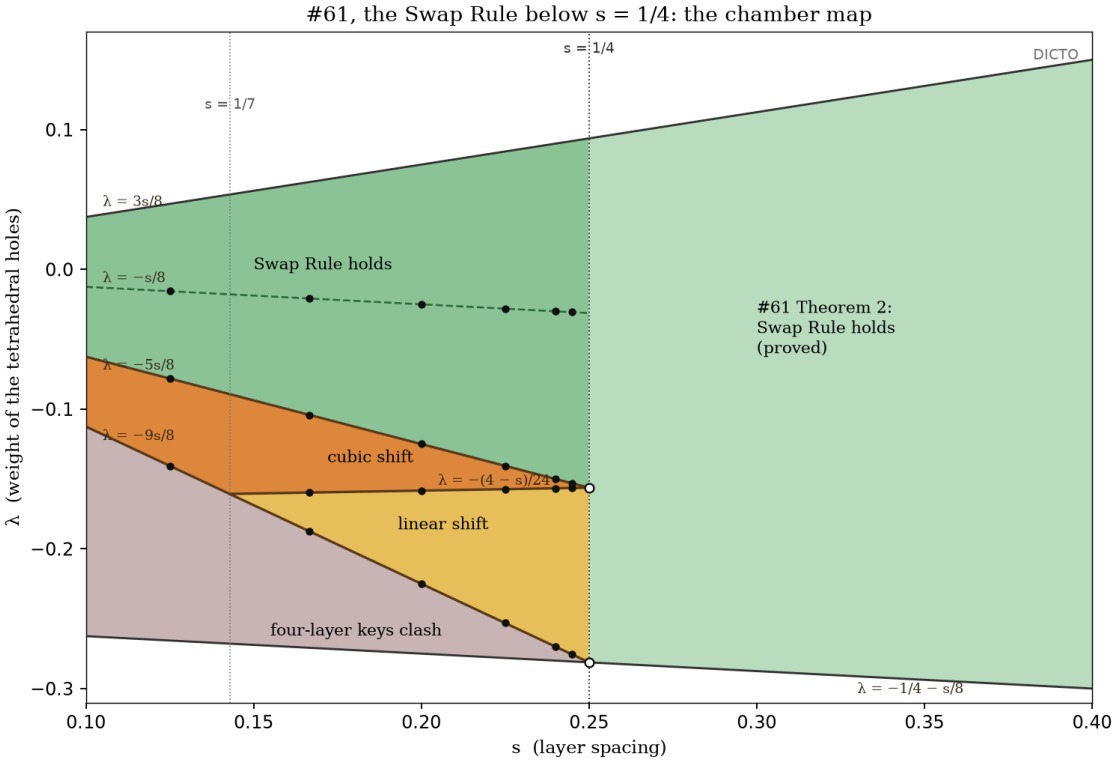


Figure 4. The chamber map of the Swap Rule below s = 1/4. Grey: four-layer keys clash; yellow: linear shift; orange: cubic shift; green: the Swap Rule holds; pale green, s ≥ 1/4: #61's Theorem 2. All cells close at s = 1/4; the walls are proved cell boundaries for every s in [1/8, 1/4] (the dots mark where they were first found).

What happens at the walls. The λ-stack of the 3D power diagrams is a 4D cell complex; in a generic weight walk its 3D slices change by the 3D T1 moves of Theorem 1′ for d = 5, one contact gained or one lost at a time. At #61's walls the lattice's central symmetry forces a loss and a gain onto one point: six cells meet there and the four exchanging holes lie in one plane with a common midpoint (a degenerate 4-4 flip, the rosette kind of event, not Theorem 1's pure swap); in that plane the exchange is a 2D scutoid swap. At s = 1/5, summed over the six words' unit cells, the faces exchanged are: at −(4 − s)/24, 57 between octahedral holes in one layer for 57 between tetrahedral holes in one layer; at −5s/8, 21 between octahedral holes two layers apart for 21 tetrahedral; at −s/8, 126 for 126 between neighbouring layers; at −9s/8, 31 long octahedral contacts lost and 173 gained. The Swap Rule switches on at the moment the last octahedral contact across a whole layer lets go.

3. One dimension up: the 4D cells of the holes

Each hole's power cell, stacked over λ in an interval $[\lambda_a, \lambda_b]$, is a convex 4-polytope $\{(x, \lambda) : n \cdot x \le b_0 + b_1\lambda,\ \lambda_a \le \lambda \le \lambda_b\}$ whose 3D slices are the cells. We propose the name *4D scutoid* for such a 4-cell whose 3D slices change neighbours at a wall where cells stop touching and others start. At s = 1/5, across the wall λ = −1/8, 25 holes (of the six words) have 4D cells with faces that end or begin exactly at the wall: the octahedral cells' faces towards octahedral holes two layers away end there, the tetrahedral cells' new faces begin there (Figure 5).

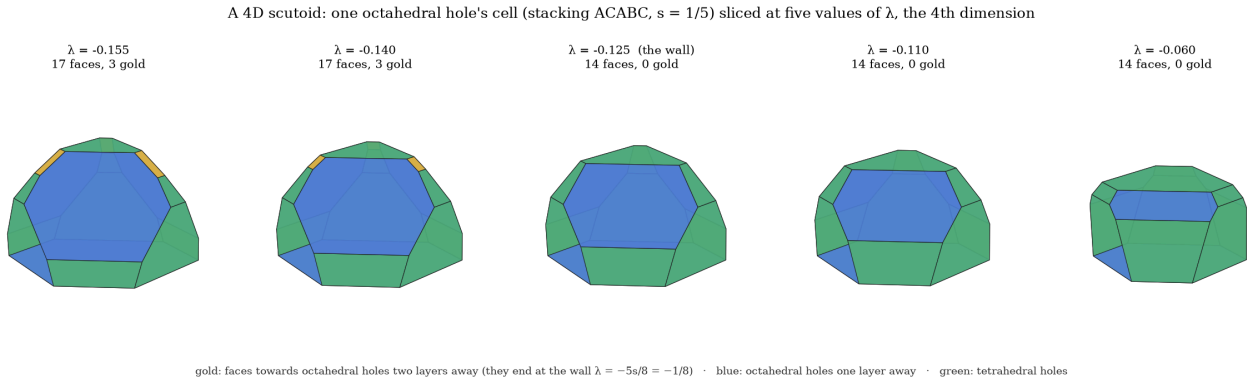


Figure 5. A 4D scutoid: one octahedral hole's cell (stacking ACABC, s = 1/5) sliced at five values of λ. Gold: faces towards octahedral holes two layers away; they end at the wall λ = −1/8 (17 faces before, 14 after).

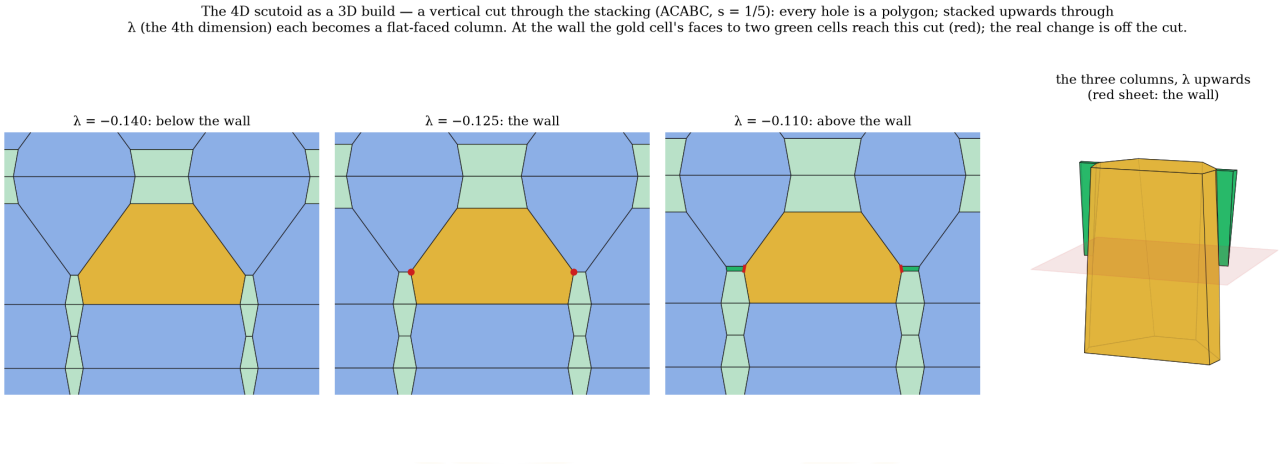


Figure 6. A vertical cut through the stacking (the plane of a set of exchanging holes), every hole a polygon, stacked through λ a flat-faced column. At the wall the two octahedral cells above and below the gold one (two layers apart) stop touching, and tetrahedral holes start touching away from this cut; the gold cell's faces towards the green cells, present throughout, reach the cut here, which is why the green columns appear as wedges at the wall.

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